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54. $m_{O_2} = 52.5$ g $\quad P_{O_2} = ?$

$m_{CO_2} = 65.1$ g $\quad P_{CO_2} = ?$

T = 300 K $\quad$ P = 9.21 atm

$P_{O_2} = X_{O_2}$ x Total Pressure

$$X_{O_2} = \frac{n_{O_2}}{n_{O_2} + n_{CO_2}}$$

$$n_{O_2} = \frac{\text{Mass of } O_2}{\text{Molar mass of } O_2}$$

$$= \frac{52.5\text{ g}}{32\text{ g mol}^{-1}} = 1.64\text{ mol}$$

$$n_{CO_2} = \frac{\text{Mass of } CO_2}{\text{Molar mass of } CO_2}$$

$$= \frac{65.1\text{ g}}{44\text{ g mol}^{-1}} = 1.48\text{ mol}$$

$$X_{O_2} = \frac{n_{O_2}}{n_{O_2} + n_{CO_2}} = \frac{1.64}{3.12} = 0.53$$

$$X_{CO_2} = \frac{n_{CO_2}}{n_{O_2} + n_{CO_2}} = \frac{1.48}{3.12} = 0.47$$

$$P_{O_2} = X_{O_2} \times \text{Total pressure}$$

$$= 0.53 \times 9.21\text{ atm} = 4.88\text{ atm}$$

$$P_{CO_2} = X_{CO_2} \times \text{Total pressure}$$

$$= 0.47 \times 9.21\text{ atm} = 4.33\text{ atm}$$

55. Pressure of the gas in the tank at its melting point

$T_1 = 298\text{ K}; P_1 = 2.98\text{ atom}; T_2 = 1100\text{ K}; P_2 = ?$

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

$$\Rightarrow \quad P_2 = \frac{P_1}{T_1} \times T_2$$

$$= \frac{2.98\text{ atm}}{298\text{ K}} \times 1100\text{ K} = 11\text{ atm}$$

At 1100 K the pressure of the gas inside the tank will become 11 atm. Given that tank can withstand a maximum pressure of 12 atm, the tank will start melting first.